Machine Learning for Experimental Design Optimisation

Conditional Neural Bayes Ratio Estimation (cNBRE)

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Why is experimental design still decided like this?

"How long should this cable be?"

Many design decisions in experimental physics are set by habit, catalogue availability, or a very small parameter sweep.

In sensitivity-limited experiments, these choices can determine whether the signal is detected at all.

This talk: use machine learning to simulate, forecast, and optimise *before* building.

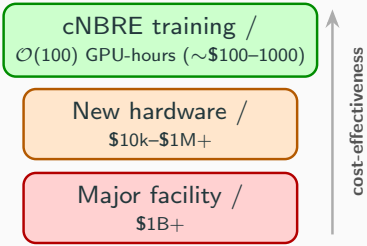
What actually needs optimising?

These are all **design parameters** – they impact the effectiveness of our system:

- Observation time: τ
- Antenna orientation: ψ
- Antenna geometry and placement
- Cable lengths and calibration hardware
- Component tolerances

Main teaser

Antenna orientation alone changes forecast detection probability by about **20 percentage points** — and costs **nothing** to implement.



Increase sensitivity for a fraction of the cost of new hardware.

Outline

Method

Results

Applications

Method

What quantity do we need for detection?

We quantify confidence in a detection by comparing the Bayesian evidence for a signal model versus a null model:

$$M_1 : \text{signal} + \text{foreground} + \text{noise}$$
$$M_0 : \text{foreground} + \text{noise}$$

$$K(\mathbf{d}) = \frac{p(\mathbf{d} \mid M_1)}{p(\mathbf{d} \mid M_0)}.$$

- $K > 1$: data prefer the signal model.
- $K \gg 1$: strong evidence for the signal.
- If I can compute K, I can tell which design gives the best chances of detection.

The bottleneck

Optimising a design parameter requires many samples over the full design space.

Each sample is an independent sampling run.

100 design choices $\times$ 100 possible signals = **10,000** expensive runs.

Add one more parameter and it becomes **1,000,000**.

How do evidence networks avoid that bottleneck?

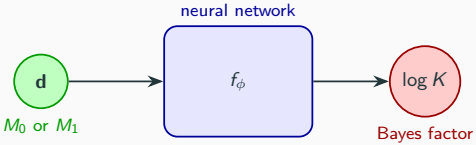
If we have a simulator for M_1 and M_0, we can train a classifier $f_\phi(\mathbf{d})$ to distinguish them.

With an appropriately chosen loss function, the network converges on the Bayes factor:

$$\mathcal{L} = \mathbb{E}_{M_1}\left[e^{-f_\phi(\mathbf{d})/2}\right] + \mathbb{E}_{M_0}\left[e^{+f_\phi(\mathbf{d})/2}\right].$$

$$f_\phi^\star(\mathbf{d}) = \log K(\mathbf{d}).$$

- Feed the network different data, it outputs the Bayes factor.
- Train once, evaluate in milliseconds.



Optimised f_ϕ^* predicts $\log K$ from data $\mathbf{d}$.

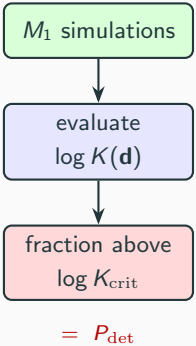
How does this become forecasting?

Given some range of physics parameters, what is our probability of detecting the signal?

Gessey-Jones et al.'s key insight was to use the Bayes factor as a forecasting utility:

$$P_{\mathrm{det}} = \Pr\left[\log K(\mathbf{d}) > \log K_{\mathrm{crit}} \,\middle|\, M_1\right].$$

- Simulate many realisations under M_1.
- Evaluate $\log K$ for each one.
- Count the fraction that cross the threshold.

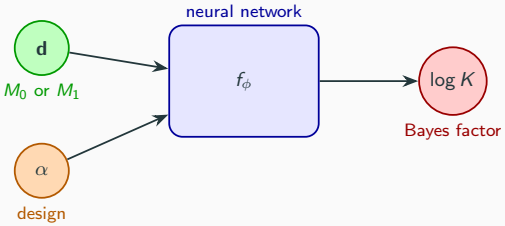


What changes in conditional neural Bayes ratio estimation?

Extend the classifier: predict which model the data came from **given some design parameter**:

$$f_\phi(\mathbf{d}, \alpha) \approx \log K(\mathbf{d} \mid \alpha).$$

- e.g. "did this data come from the signal model, **given some cable length**?"
- Train once across the full design space.
- Query **any** α in milliseconds.



Key idea

Amortised: check how ANY design choice impacts detection odds.

Application to 21-cm cosmology with REACH

A useful testing ground: tough, unsolved problem with dim signal compared to foreground and very advanced, fast simulators.

- Signal amplitude: about **100 mK**
- Foregrounds: 10^4–10^5 **times brighter**
- Forward model: physics-based and implemented in JAX

If cNBRE works here, it is worth taking seriously elsewhere.

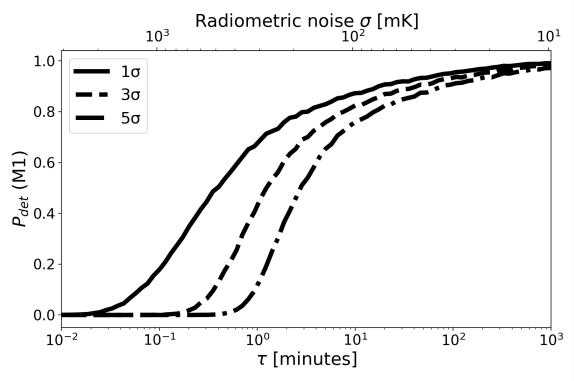
$$M_1 : \mathbf{d} = \mathbf{d}_{\mathrm{fg}} + \mathbf{d}_{21} + \mathbf{n}$$

$$M_0 : \mathbf{d} = \mathbf{d}_{\mathrm{fg}} + \mathbf{n}$$

REACH

Results

Does cNBRE recover the right physics?



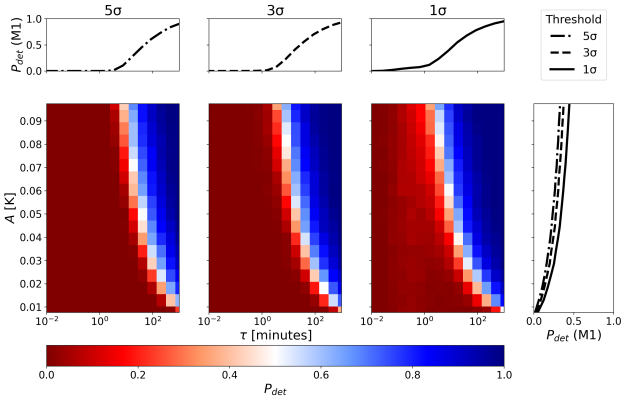
We start on a simple problem where we know the answer: observation time as the design parameter.

- As τ increases, noise goes down, so detection should get easier.
- The trained network tells us: "your probability of detecting the signal at this observation time is X".
- The network learns the radiometer-equation scaling.

Validation

The network reproduces the basic physical behaviour we expect. It does.

Amortised: amplitude × observation time

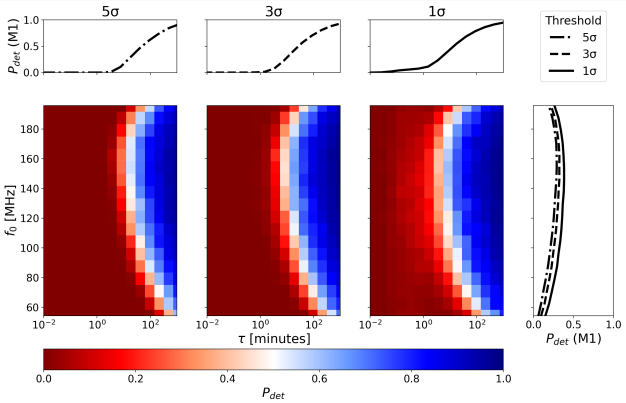


Amortised exploration

Given some range of **physical signal parameters**, what is our P_{det} given our **design parameter**? **Same network, no retraining.**

- Each pixel would have required a separate nested sampling run.
- The joint map shows how instrument choices and the underlying physics interact.
- Ask about amplitudes today, signal *locations* tomorrow – just re-slice.

Amortised: frequency × observation time



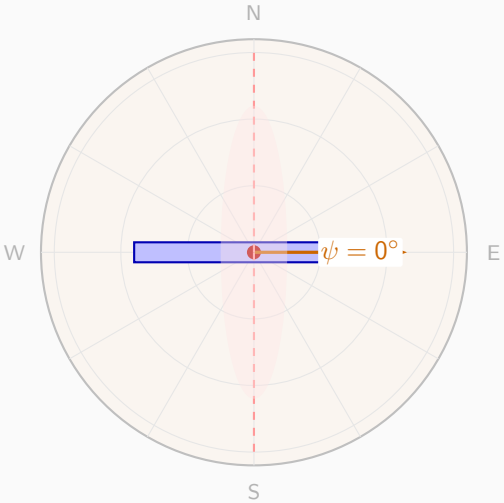
Same network, different slice

Answer a new question without re-training, in less than 1 second.

Signals at 100–140 MHz are easier to detect; sensitivity drops at band edges.

- This is the **same trained network** as the previous slide.
- We just sliced down a different axis of the detection landscape.

How should we orient our antenna?



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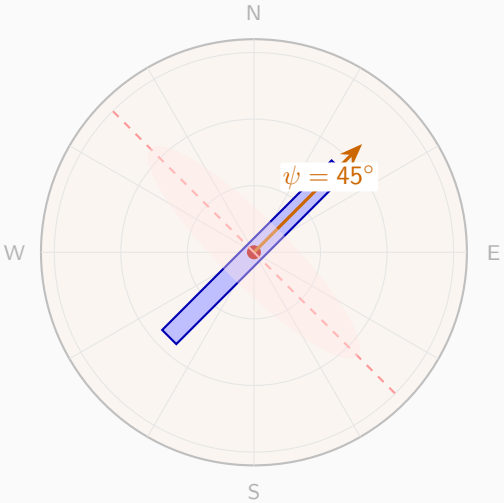

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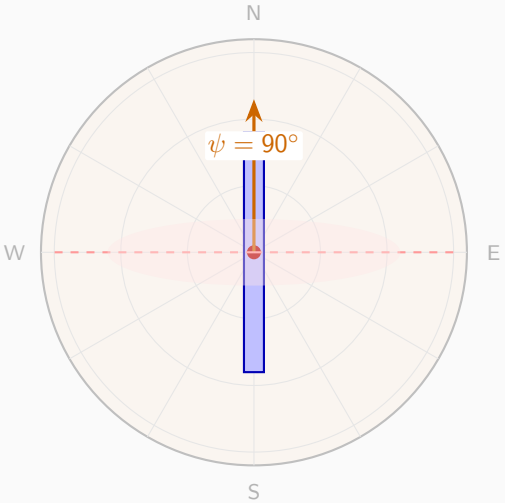

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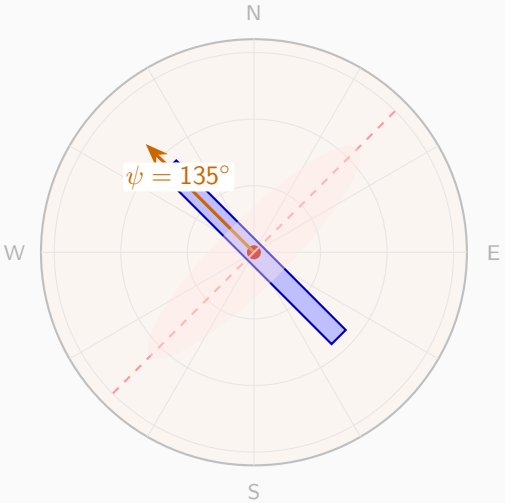

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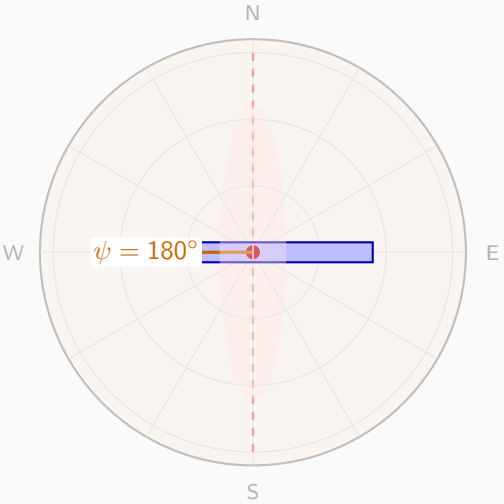

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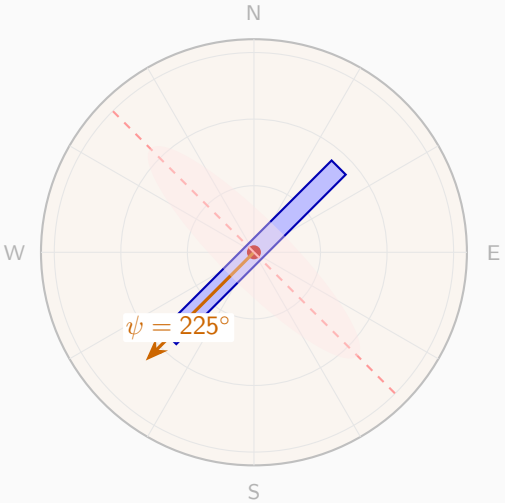

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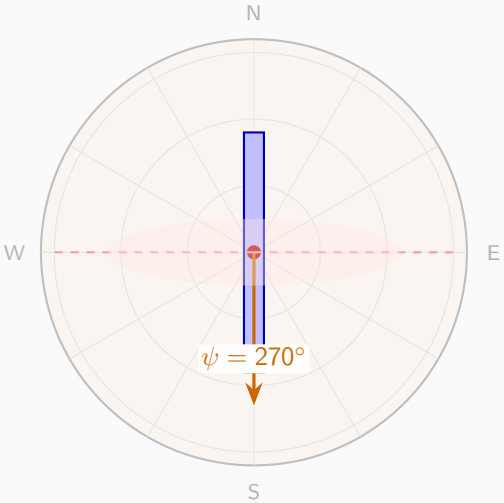

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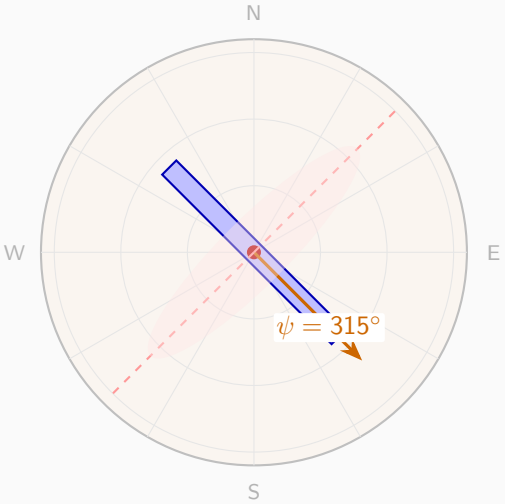
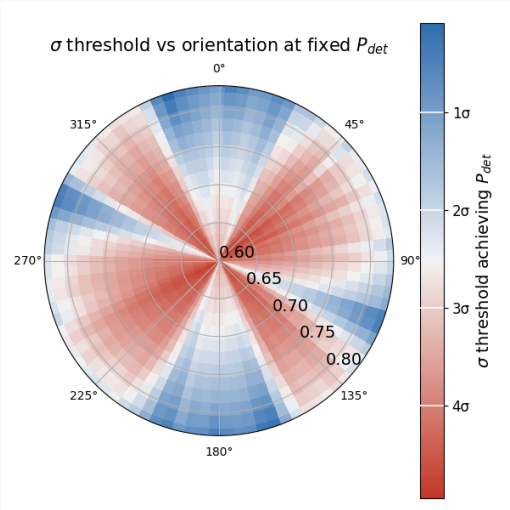


Can one design choice really move the needle that much?



One night of observation on REACH:

- We vary only antenna orientation.
- Detection probability changes by about **20 percentage points**.
- Four-fold symmetry from dipole beam geometry.

Why this matters

This costs **nothing**. How much would the funders of any state-of-the-art experiment pay for a 20% increase in sensitivity?

Applications

What do you need to use cNBRE?

All you need is a **forward simulator** and a **design parameter** you want to optimise.

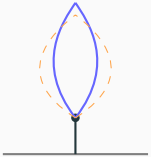
Most experiments that do parameter estimation **already have** a forward model inside their likelihood.

1. Define your two models (M_1 vs M_0).
2. Identify your design parameters α.
3. Simulate training data across the design space.
4. Train a single cNBRE network.

The simulator does not need to be differentiable, fast, or GPU-based.

It just needs to produce realistic mock data for any α.

Instrument design



Antenna geometry

$\alpha =$ beam parameters

Optimise: P_{det}

Calibration hardware

$\alpha = \ell$ (cable length)

Optimise: P_{det}

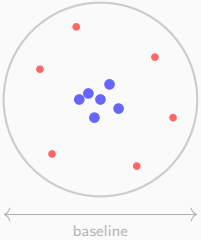
Component tolerance

$\alpha = \delta_{\mathrm{VNA}}, \delta_{\mathrm{LNA}}, \ldots$

Optimise: P_{det}

These parameters are fixed **before construction**. Getting them wrong is expensive to undo. cNBRE lets you explore the full design space from a single training run.

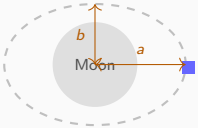
Array and orbit design



Array configuration

$\alpha =$ baseline distribution

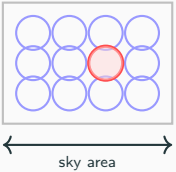
Optimise: P_{det}



Lunar orbit

$\alpha = (a, b, i, \ldots)$

Optimise: P_{det}



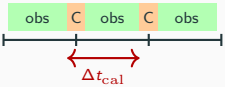
Integration per pointing

$\alpha = t_{\mathrm{point}}$

Optimise: P_{det}

Array layout, orbit parameters, and survey strategy all involve high-dimensional trade-offs where exhaustive grid search is intractable.

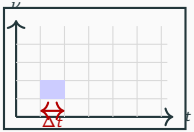
Operations and scheduling



Calibration cadence

$\alpha = \Delta t_{\mathrm{cal}}$

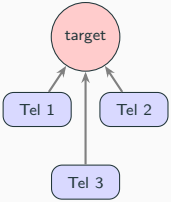
Optimise: P_{det}



Correlator integration

$\alpha = \Delta t_{\mathrm{dump}}$

Optimise: P_{det}



Multi-telescope scheduling

$\alpha =$ time allocation

Optimise: P_{det}

Operational parameters can be re-optimised throughout the lifetime of the experiment – cNBRE amortises the cost of exploring new strategies.

Key takeaways

1. **Optimise one or several design parameters simultaneously.**

2. Combine design optimisation with physical models of the signal you are searching for.

3. Significantly increases the chance of achieving science goals, and is highly cost-effective.

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Thanks for listening!

Questions?

Slides and source:

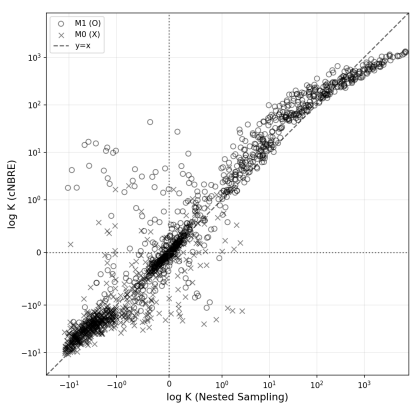
github.com/samleeney/Talks

Email:

sakl2@cam.ac.uk

QR code to the talk repository

Does the amortised answer agree with nested sampling?



- cNBRE broadly tracks nested sampling on held-out simulations.
- For null-model data, cNBRE is more conservative: fewer false positives.
- Training: $\mathcal{O}(100)$ **GPU-hours once**. Evaluation: **milliseconds**.

Nested sampling remains the reference calculation.

cNBRE matters because it amortises the expensive step.

Derivation: optimal network output is the log-Bayes factor

Minimise the expected exponential loss:

$$\mathcal{J}[f] = \int \left[p(\mathbf{d}|M_1)\, e^{-f(\mathbf{d})/2} + p(\mathbf{d}|M_0)\, e^{f(\mathbf{d})/2} \right] \mathrm{d}\mathbf{d}.$$

Set the functional derivative to zero:

$$\frac{\delta \mathcal{J}}{\delta f} = -\tfrac{1}{2}\, p(\mathbf{d}|M_1)\, e^{-f^*/2} + \tfrac{1}{2}\, p(\mathbf{d}|M_0)\, e^{f^*/2} = 0.$$

Rearranging:

$$e^{f^*(\mathbf{d})} = \frac{p(\mathbf{d}|M_1)}{p(\mathbf{d}|M_0)} = K(\mathbf{d}) \qquad \Longrightarrow \qquad \boxed{f^*(\mathbf{d}) = \log K(\mathbf{d})}.$$

Conditioning on a design parameter α: replace $p(\mathbf{d}|M_i) \to p(\mathbf{d}|M_i, \alpha)$ throughout. The same derivation gives

$$\boxed{f^*(\mathbf{d}, \alpha) = \log K(\mathbf{d} \mid \alpha) = \log \frac{p(\mathbf{d}|M_1, \alpha)}{p(\mathbf{d}|M_0, \alpha)}}.$$